

Monte Carlo Pricing of Asian Options

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1 Setup

Take S_t to follow geometric Brownian motion under the risk-neutral measure $\mathbb{Q}$:

$$dS_t = rS_t dt + \sigma S_t dW_t, \quad (1)$$

with r the risk-free rate, σ the volatility and W_t a standard Brownian motion. The solution is

$$S_T = S_0 \exp\left(\left(r - \frac{1}{2}\sigma^2\right) T + \sigma\sqrt{T} Z\right), \quad Z \sim \mathcal{N}(0, 1), \quad (2)$$

and equation (2) is what the simulation samples from directly.

2 European call as a control

The European call has a closed form,

$$C = S_0\Phi(d_1) - Ke^{-rT}\Phi(d_2), \quad d_{1,2} = \frac{\ln(S_0/K) + \left(r \pm \frac{1}{2}\sigma^2\right) T}{\sigma\sqrt{T}}, \quad (3)$$

where Φ is the standard normal CDF. Since the answer is already known, pricing it by simulation is a check on the estimator. If the simulated price sits inside the confidence interval around the analytic price, the sampling and discounting are behaving.

3 Asian call

An arithmetic-average Asian call pays

$$\max\left(\frac{1}{n} \sum_{i=1}^n S_{t_i} - K, 0\right). \quad (4)$$

A sum of lognormal variables is not lognormal, so there is no closed form to fall back on. The estimator is

$$C_{\text{Asian}} = e^{-rT} \mathbb{E}^{\mathbb{Q}}[\max(\bar{S} - K, 0)] \approx \frac{e^{-rT}}{M} \sum_{j=1}^M \max(\bar{S}^{(j)} - K, 0) \quad (5)$$

over M paths. Convergence is $O(M^{-1/2})$, so accuracy is bought expensively, and cutting the error in half costs four times the paths.

4 Results

With $S_0 = K = 100$, $r = 5\%$, $\sigma = 20\%$ and $T = 1$:

	Price	95% interval
European, analytic	10.4506	n/a
European, Monte Carlo	10.4520	± 0.0204
Asian, Monte Carlo	5.7661	± 0.0350

The simulated European price differs from the analytic one by 0.0014, well inside the interval, so the estimator is sound. The Asian call prices at roughly half the European. Averaging over the path damps the variance of the terminal quantity, and a less volatile underlying quantity carries less optionality.